

Deep brain stimulation for obsessive-compulsive disorder: evolution of tractography-based targeting

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OBJECTIVE Deep brain stimulation (DBS) in the anterior limb of the internal capsule (ALIC) for obsessive-compulsive disorder (OCD) can result in large improvements in symptoms and quality of life. However, without a consistent stimulation target within the anatomically variable ALIC region, the therapy has required long and unpredictable trial-and-error periods of parameter optimization to achieve these clinically meaningful benefits. To facilitate scalability and clinical implementation of ALIC DBS for OCD, this study aimed to demonstrate the evolution of prospective DBS targeting using patient-specific tractography.

METHODS Initially, the authors targeted a white matter map of ALIC bundles and cortical regions previously implied in DBS for OCD. They then generated a map of the ALIC connection shared by 6 participants with OCD who responded to DBS. This common responder map implicated white matter connections to the ventromedial prefrontal/orbitofrontal cortex (vmPFC/OFC), ventrolateral prefrontal cortex (vlPFC), and midbrain. Next, we prospectively targeted a new series of 7 patients guided by this common map. Finally, we generated a tractography-based predictive model based on individual stimulation-response outcomes to estimate the relative importance of each targeted ALIC pathway.

RESULTS Targeting a sweet spot in the common responder ALIC map that optimally connected the patient-specific vmPFC/OFC, vlPFC, and midbrain led to a consistent and predictable reduction of OCD symptoms, with 8 of 10 patients attaining a clinically meaningful Yale-Brown Obsessive-Compulsive Scale (Y-BOCS) score reduction in symptom severity of at least 35%. Individualized tractography-based targeting improved the primary OCD symptoms independent of hypomania or other changes in mood or anxiety and required minimal parameter changes away from the empirical tractography-based target spot. The tractography-based stimulation model generated a connectivity profile of left and right ALIC pathways that selectively predicted Y-BOCS score reduction and could be used to optimize future targeting further.

CONCLUSIONS The authors' results show how patient-specific tractography can facilitate clinically effective precision targeting in ALIC DBS for OCD.

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KEYWORDS DBS; deep brain stimulation; OCD; obsessive-compulsive disorder; ALIC; anterior limb of the internal capsule tractography; targeting; common map; functional neurosurgery

ABBREVIATIONS ALIC = anterior limb of the internal capsule; BAI = Beck Anxiety Inventory; BDI = Beck Depression Inventory; BEDPOSTX = Bayesian Estimation of Diffusion Parameters Obtained Using Sampling Techniques; dACC = dorsal anterior cingulate cortex; DBS = deep brain stimulation; dlPFC = dorsolateral PFC; dmPFC = dorsomedial PFC; dMRI = diffusion-weighted MRI; FBC = fiber bundle capacity; GPe = globus pallidus pars externa; MSE = mean squared error; OCD = obsessive-compulsive disorder; PFC = prefrontal cortex; ROI = region of interest; STN = subthalamic nucleus; vACC = ventral anterior cingulate cortex; vlPFC = ventrolateral PFC; vmPFC/OFC = ventromedial PFC/orbitofrontal cortex; Y-BOCS = Yale-Brown Obsessive-Compulsive Scale.

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OBSESSIVE-COMPULSIVE disorder (OCD) is a neuropsychiatric disorder affecting 2%–3% of the population.^{1–3} OCD is characterized by persistent obsessions and compulsions causing significant distress and impairment.^{2,4} Conventional treatments include pharmacotherapy and cognitive-behavioral therapy, but 20%–30% of OCD patients develop severe refractory symptoms.^{1,2}

Deep brain stimulation (DBS) is an effective neuromodulation treatment for patients with refractory OCD. Studies across different anatomical targets in the striatal region^{2,3} have reported highly promising though variable outcomes with response rates between 40% and 70% and symptom decreases between 30% and 60% in more than 440 patients with OCD.^{3,5} The anterior limb of the internal capsule (ALIC) is currently the most widely used and only FDA-approved DBS target for OCD (as a Humanitarian Device Exemption). ALIC DBS has variable response rates between 20% and 70% in an estimated total of 270 patients and requires lengthy trial-and-error parameter optimizations.^{1,6} These constraints have likely contributed to its limited scalability and clinical implementation and reflect an incomplete knowledge of the optimal anatomical target within the ALIC region.

The ALIC aggregates white matter tracts implicated in a range of affective and cognitive functions underlying OCD and depression, such as emotion regulation and controlled decision-making.⁷ Anatomical and tractography approaches have studied the complex ALIC structure to help delineate therapeutic connections in OCD DBS. These studies have suggested a ventrodorsal and mediolateral topological organization of ALIC fibers to prefrontal cortex (PFC) regions^{8,9} and a medial organization of fibers to the thalamus and lateral to the subthalamic nucleus (STN).¹⁰ Normative connectome analyses from multiple independent centers have suggested that OCD improvement may correlate with stimulation of ALIC connections between the lateral and medial PFC, thalamus, and midbrain.^{11–14} Targeting only the lateral ALIC fibers to the ventral tegmental area modestly improved the response rate with fewer side effects.¹⁵ Recently, aggregated retrospective multicenter data suggested specific improvement in OCD symptoms at stimulation sites that, relative to the anterior commissure, are located either anterodorsally in the central ALIC or posteroventrally in the region of the inferior thalamic peduncle and bed nucleus of the stria terminalis.¹⁶

However, a map of therapeutic ALIC connections that can be implemented for prospective surgical targeting in OCD DBS and achieve predictable positive outcomes is still lacking. In DBS for major depressive disorder, inconsistent targeting also posed challenges.¹⁷ Systematic mapping of white matter connections to the subcallosal cingulate DBS target in patients with an antidepressant response resulted in a clear four-bundle common responder map used for DBS targeting. This common map method improved response rates with minimal parameter changes^{18–20} and facilitated the development of an electrophysiological biomarker of antidepressant response,²⁰ motivating us to develop a similar precision method for OCD.

Here, we report our evolution toward developing a tractography-based common map approach for precision targeting ALIC DBS in OCD in a series of 13 consecutive

patients. We hypothesized that our tractography-based targeting method would result in a more uniform target location and fewer trial-and-error parameter adjustments to achieve the best outcome, benefiting future clinical implementation of DBS for OCD. First, we generated a common responder map of white matter ALIC connections shared by all OCD DBS responders. Next, we used this map for prospective targeting in a new cohort of patients. Finally, we generated a tractography-based stimulation model for retrospectively predicting clinical outcomes based on selective left and right ALIC pathways that can be used for future testing of optimized targeting.

Methods

Participants

We included 13 patients diagnosed with severe OCD who consecutively underwent implantation (2019–2023) with bilateral DBS leads in the ALIC using tractography-based targeting at Mount Sinai Hospital. Detailed inclusion criteria and a summary of patient demographic characteristics (Supplementary Table 1) are available in the Supplementary Methods.

We divided this series of 13 patients into 3 cohorts that reflect the investigational evolution of our tractography-based targeting method. Our first two cohorts were used to develop our initial tractography common responder map, which we then tested for prospective targeting in our third cohort. Patients in cohort 1 ($n = 3$) underwent implantation between March 2019 and September 2020 with Abbott-St. Jude 6173 directional leads with 1.5-mm electrode spacing (St. Jude Medical). In this first cohort, in the absence of a tractography-based common responder map for OCD DBS targeting, we chose to implant DBS leads using patient-specific tractography to target the medial ALIC (anterior thalamic radiation) and lateral ALIC (white matter bundles connected to the ventral tegmental area, medial forebrain bundle). Our rationale for targeting these specific ALIC projections was based on previous retrospective tractography analyses of DBS for OCD, suggesting improved response with stimulation of medial ALIC to thalamus²¹ or lateral ALIC to brainstem.¹¹ This first cohort of patients were all responders at the last available follow-up assessment (31–36 months post-DBS). We used only the tractography data of these 3 responders to generate the common responder map but not their symptom scores, as these were not systematically collected for all time points of the 2 prospective cohorts.

Cohorts 2 ($n = 3$) and 3 ($n = 7$) consisted of consecutive patients enrolled in a prospective longitudinal study aimed at investigating tractography-based DBS targeting for OCD (ClinicalTrials.gov identifier NCT05160129). Medtronic SenSight B33015 directional leads with 1.5-mm electrode spacing (Medtronic) were implanted. Patients in cohort 2 underwent implantation between December 2021 and March 2022. An additional patient who underwent implantation during this period was excluded from the study due to lead explanation following an infection. From cohort 1 to 2, our tractography-based targeting method evolved from only including medial and lateral ALIC to also including cortical ALIC projections as

identified by a new anatomical study by Haber et al., i.e., dorsomedial PFC (dmPFC), dorsolateral PFC (dlPFC), ventrolateral PFC (vlPFC), ventromedial PFC (vmPFC)/orbitofrontal cortex (OFC), and dorsal anterior cingulate cortex (dACC).⁹ The rationale for additionally targeting these cortical regions was based on their association with OCD or OCD-relevant cognitive functions⁷ and because they can be easily missed with human diffusion MRI of the ALIC but were strongly implicated in Haber's nonhuman high-resolution tract tracing and diffusion imaging results.⁹ Finally, we used tractography data from cohorts 1 and 2 to develop an initial common responder map used for prospective targeting in cohort 3 (see *Initial Common Responder Map* in *Methods*). Patients in cohort 3 underwent implantation between August 2022 and June 2023.

In cohorts 2 and 3, symptoms were systematically assessed using the Yale-Brown Obsessive-Compulsive Scale (Y-BOCS), Beck Depression Inventory (BDI), Beck Anxiety Inventory (BAI), and Young Mania Rating Scale at presurgical baseline, postsurgically prior to DBS turn-on, weekly during the first 4 weeks after DBS turn-on, twice a month for a period of 12 months, and every 2 months after 12 months. Clinical assessments were collected by a trained rater, and the stimulation settings for each session were recorded.

Image Acquisition

The preoperative imaging protocol included a structural T1-weighted and diffusion-weighted MRI (dMRI) acquired using a 3T GE Signa Architect scanner (GE Healthcare). Additionally, for patients in cohorts 2 and 3, multishell high spatial resolution dMRI data in a 7T Siemens scanner were acquired. See Supplementary Methods for detailed MRI parameters. Preprocessing steps of the dMRI data included denoising and removal of Gibbs's ringing artifacts in MRtrix3 and susceptibility-induced distortions and eddy current correction with FSL (FMRIB's Software Library) topup and eddy, respectively.

Lead Reconstruction and Stimulation Volumes

We performed lead localization using the Lead-DBS toolbox²² in MATLAB 2022b (Fig. 1A). DBS leads were semiautomatically localized with the PaCER (Precise and Convenient Electrode Reconstruction for Deep Brain Stimulation) algorithm with a postoperative CT scan, and the segmented lead's orientation was matched to the implantation. Stimulation volumes were estimated using the SimBio/Fieldtrip pipeline in Lead-DBS with default tissue conductivity and electric field threshold values. The stimulation volumes corresponding to the last DBS programming visit for each cohort were aggregated to generate an N-image or heat map.²³ The values in these images represent the number of times each voxel was activated across the stimulation volumes (i.e., voxels with higher values were activated across more patients).

Initial Common Responder Map

We generated our initial tractography common responder map using similar methods described by Riva-Posse et al. for subcallosal cingulate DBS.¹⁹ In brief, our initial

common responder map was based on 3T dMRI and clinical data from cohorts 1 and 2, of which all 6 patients were responders at the last follow-up (Fig. 1B). We generated stimulation volumes for each patient using their DBS parameters at the time of response (i.e., 4.5–6.5 mA, 130 Hz, 90 μ sec). We performed probabilistic tractography from the patient-specific stimulation volumes as seed regions to the rest of the brain using FSL probtrackx2 followed by BEDPOSTX (Bayesian Estimation of Diffusion Parameters Obtained Using Sampling Techniques). We generated 5000 streamlines per seed voxel using a CSF avoidance mask. To calculate the structural connection probability, the whole-brain tractography map was divided by the total number of survived streamlines (i.e., waytotal). Finally, individual white matter structural connectivity maps were warped into Montreal Neurological Institute (MNI) space, with a threshold at 0.1% to remove false positives, and then averaged. Voxels shared by all 6 responders are shown in the initial common responder map.

Tractography-Based Surgical Targeting

In cohort 1, we first identified 2 previously reported white matter tracts (anterior thalamic radiation: medial ALIC; and medial forebrain bundle: lateral ALIC connected to the ventral tegmental area) associated with clinical improvement.^{11,21} We then selected the overlap region between these 2 tracts within the ALIC as the target location of the active DBS contact.

In cohort 2, our targeting method evolved from a location connecting lateral and medial ALIC to one also connecting specific cortical ALIC projections (dmPFC, dlPFC, vlPFC, vmPFC/OFC, and dACC). We parcellated the ALIC into 5 subregions using the probability of connections from the ALIC to each cortical target. The target location was then selected to maximally involve vmPFC/OFC and vlPFC projections since these two cortical projections were commonly shown in the responders of cohort 1.

Finally, we used our initial responder common map for patient-specific tractography-based targeting in cohort 3. At the cortical level, the initial common responder map (Fig. 2A) consisted of connections to the vmPFC/OFC and vlPFC. At the subcortical level, responders shared a midbrain pathway with two "subbranches": one lateral to the red nucleus and medial to the STN and a second one medial to the red nucleus. We determined our target location ("sweet spot") to maximally stimulate the connections from the common map (i.e., the 3 ALIC pathways to vmPFC/OFC, vlPFC, and lateral and medial midbrain) (Fig. 2B). We implanted the DBS leads so the third tier of contacts from the bottom lead tip would overlap with this sweet spot, allowing for the potential activation of neighboring contacts within the common map. The distal or anchor point of the DBS lead was placed within the anteromedial globus pallidus pars externa (GPe). The exact location in relationship to the external borders or the GPe/globus pallidus pars interna border was variable and dependent on a given patient's specific white matter anatomy.

Tractography-Based Predictive Model

We used data from cohorts 2 and 3 to generate a trac-

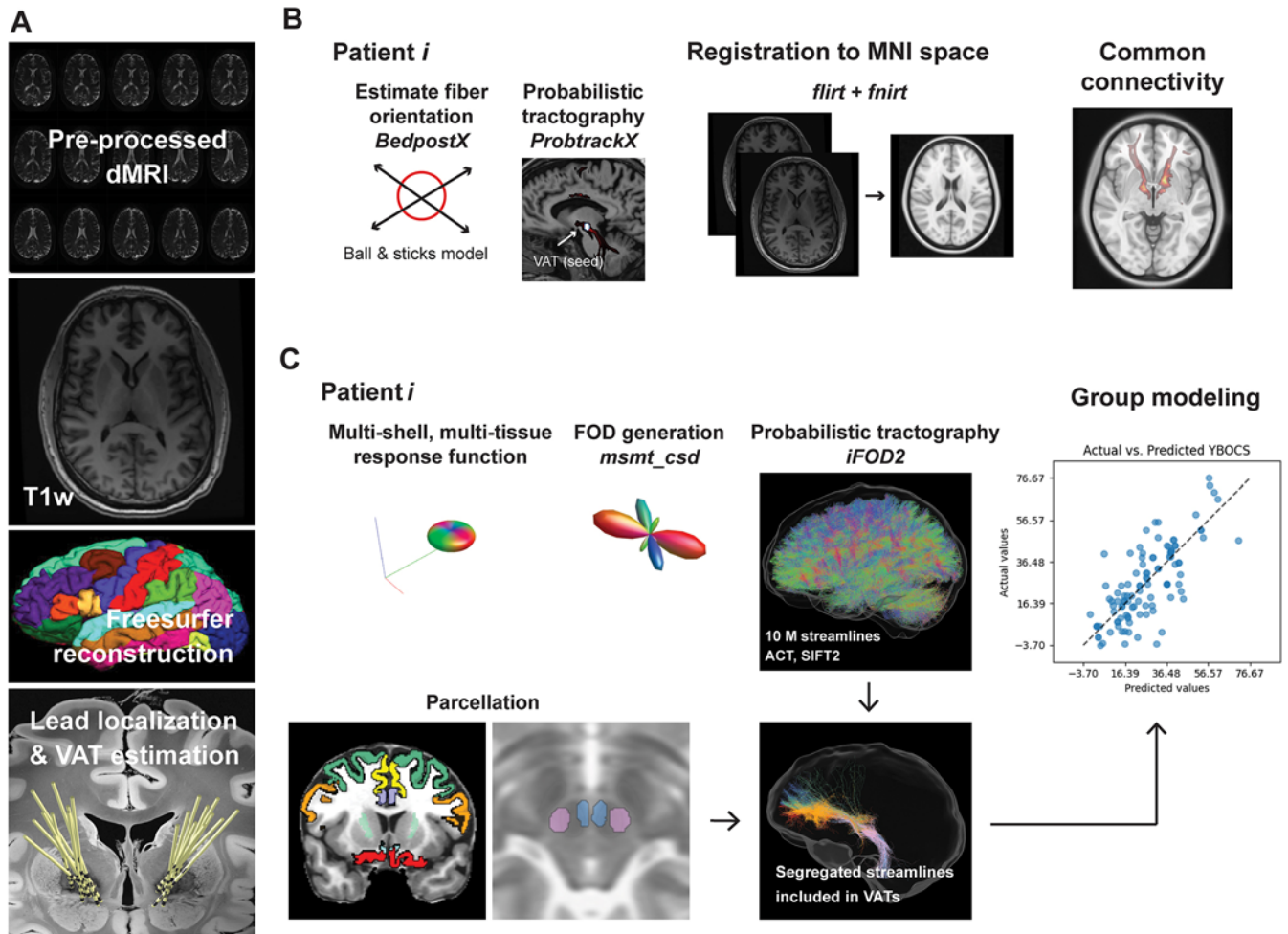


FIG. 1. Methodology for the tractography analysis. **A:** Input data for panels B and C consists of the preprocessed dMRI, T1-weighted (T1w) structural scans, Freesurfer cortical reconstruction, lead localization, and generation of stimulation volumes (volume of activated tissue [VAT]). **B:** Methodology to generate the common map of responder patients (cohorts 1 and 2). For each patient, streamlines were generated from the stimulation volumes (VATs) to the rest of the brain. The individual streamlines were warped into MNI space and averaged to generate a responder common connectivity map. **C:** Methodology to generate the tractography-based stimulation model in ALIC DBS (cohorts 2 and 3). For each patient, a whole-brain tractogram was generated. The cortical and subcortical parcellation was used to segregate the streamlines from the whole-brain tractogram included in the stimulation volumes (VATs). A linear model that related the individual pathway activations to the clinical scores was done to predict clinical improvement. Figure is available in color online only.

tractography-based stimulation model predicting the therapeutic importance of individual ALIC DBS connections (Fig. 1C). For each patient, we performed probabilistic tractography in MRtrix3. First, we estimated the individual multi-shell multitissue response functions. Then, we calculated the population-averaged response functions to obtain the patient-specific orientation distribution. After performing intensity normalization, we generated a whole-brain tractogram of 10 million streamlines with the iFOD2 algorithm and anatomically constrained tractography. We addressed the nonquantitative nature of streamline counts with the SIFT2 algorithm.²⁴ This algorithm assigns weights to individual streamlines of a whole-brain tractogram to quantitatively estimate white matter connectivity. We used the fiber bundle capacity (FBC) to compare the results across patients. This quantitative metric estimates the bandwidth

of a white matter pathway, and it is obtained from the combination of the reconstructed tractogram with fiber density information derived from the diffusion imaging data.²⁵

For each patient, we quantified the connectivity of streamlines (FBC) included in their ALIC stimulation volume. Then, we segregated the streamlines according to their connections with parcellated cortical and subcortical regions of interest (ROIs) (Fig. 3). ROIs were derived from previous studies describing ALIC connectivity:^{8,9} dlPFC, dmPFC, vlPFC, vmPFC/OFC, dACC, ventral anterior cingulate cortex (vACC), paracingulate cortex, thalamus, and STN. Two additional brainstem ROIs were empirically derived from our initial common responder map: a lateral midbrain region connected to the lateral midbrain responder tract lateral to the red nucleus and medial to the STN, and a medial midbrain region connected to the

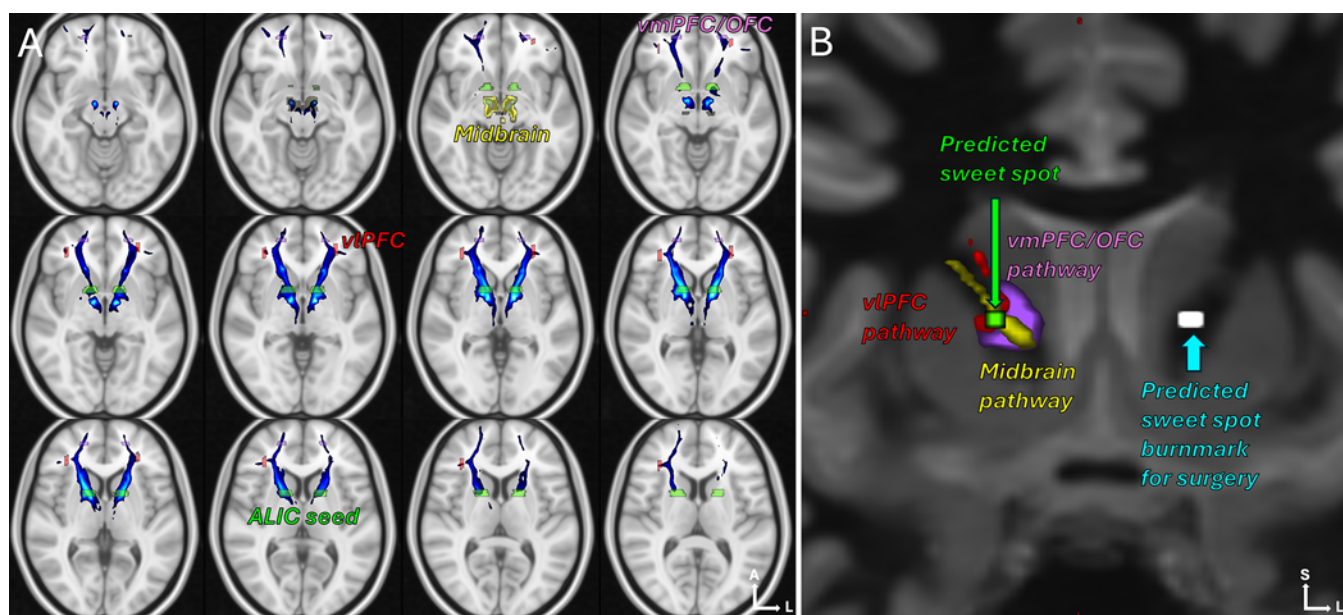


FIG. 2. Initial common responder map. **A:** Probabilistic tractography map of ALIC DBS connections shared by all 6 OCD responders (cohorts 1 and 2). The responder map (blue) consists of 3 white matter pathways to the vIPFC (red), vmPFC/OFC (purple), and midbrain (yellow). Brighter colors in the heat map represent a higher number of streamlines. **B:** We defined the surgical target (sweet spot, green) for new patients (cohort 3) based on the highest connectivity to these 3 seed regions and marked the sweet spot (white) on the anatomical image for surgery. Figure is available in color online only.

medial midbrain responder tract medial to the red nucleus (Fig. 3B). For a detailed description of the areas included in the parcellation, see Supplementary Methods and Supplementary Table 2.

To analyze the tractography data, we first obtained the correlation between the activation of individual ALIC pathways and the clinical outcomes using repeated measures.²⁶ Second, we fitted 3 linear mixed-effect models with a random intercept to generate the tractography-based predictive models to predict Y-BOCS, BDI, and BAI score improvement. The activations of each unilateral ALIC pathway were the predictor variables, while the score's reduction from baseline was the outcome variable. We included data from 197 visits, corresponding to 64 unique DBS settings. We obtained the mean squared error (MSE) and the predictions for symptoms' scores in leave-one-subject-out cross-validation. Finally, we calculated the correlation between the improvement in empirical clinical scores and the model predictions using repeated measures.

Updated Common Responder Map

Finally, we determined a new, updated common responder map based on high-resolution 7T dMRI from the 7 responder patients from the prospective cohorts 2 and 3. After generating the whole-brain tractogram for each patient (see *Initial Common Responder Map*), we selected the streamlines included in the stimulation volumes corresponding to their last follow-up assessment. Then, we generated a tract-density image containing the information of the streamlines' weights.²⁷ Finally, we warped the individual tract-density images into MNI152 NLIN 2009b space and averaged them to generate the updated common responder map.

Results

Initial Common Responder Map

The initial common responder map (Fig. 2A) consisted of connections to the vmPFC/OFC and vIPFC at the cortical level. At the subcortical level, responders shared a midbrain pathway comprising a lateral and medial subbranch: a lateral midbrain pathway lateral to the red nucleus and medial to the STN and a medial midbrain pathway medial to the red nucleus.

Stimulation Location

DBS lead localization confirmed an evolution of precision ALIC targeting from the first two cohorts used to empirically generate a responder map to actually targeting that map in cohort 3 (Fig. 4 left). The active DBS contacts corresponding to the final stimulation settings were anatomically located closest to each other in cohort 3 (Fig. 4 right). In this cohort, the active contacts were always located at the boundary of the GPe. The mean coordinates for the active contacts are detailed in Supplementary Table 3. The aggregated stimulation volumes per cohort (i.e., N-images) showed that the activation map for cohort 3 was more medial and anterior than the activation map for cohort 2 (Fig. 5).

Clinical Outcomes

We analyzed the clinical score reduction from baseline after DBS turn-on at month 1, month 6, and the last follow-up individually per cohort and as a group (cohorts 2 and 3) (Fig. 6, Supplementary Table 4, and Supplementary Fig. 1). For cohort 1, we did not have the exact same clinical follow-up time points, so we only used

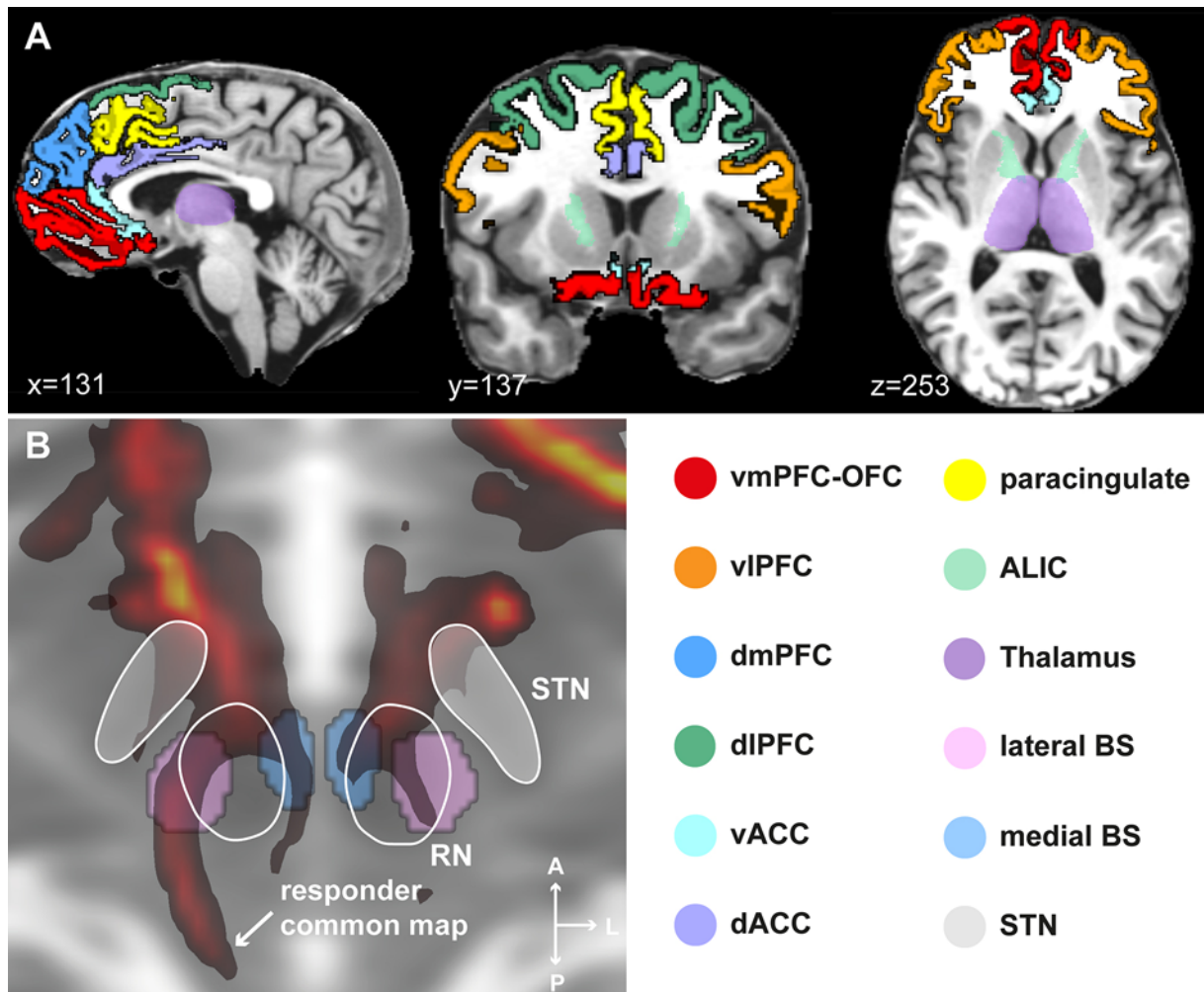


FIG. 3. Parcellation to segregate the ALIC pathways for the stimulation model. **A:** Sagittal, coronal, and axial views of the defined areas. **B:** Areas defined at the brainstem (BS) level overlaid on the top of the responder common map. Warmer colors in the responder map indicate a higher number of streamlines. RN = red nucleus. Figure is available in color online only.

their tractography data to generate the initial common responder map.

Tractography-based ALIC DBS improved OCD symptoms across the 10 prospective patients by 45.7% with an overall 80% response rate, meaning that 8 of 10 patients had a clinically meaningful Y-BOCS score reduction in symptom severity of at least 35%. Although all patients in cohort 2 but not cohort 3 were responders at the last follow-up, OCD improvement was faster and more prominent in cohort 3 using consistent targeting of the common map (month 1, median Y-BOCS improvement 7.7% in cohort 2 and 19.4% in cohort 3; month 6, 38.9% in cohort 2 and 41.2% in cohort 3). Targeting based on the common map was also followed by fewer trial-and-error parameter adjustments to achieve optimal clinical benefits. In cohort 2, the single bilateral active contact from the initial planning was changed to another contact in 2 of the 3 patients to achieve a clinical response. In this cohort, each patient required 1–4 contact changes and 6–10 amplitude changes (Supplementary Table 1). In cohort 3, 5 of 7 patients achieved clinical response with stimulation at the single contact correspond-

ing to the tractography sweet spot with only 1 to 3 amplitude changes (Supplementary Table 1). Two patients (one with comorbid autistic spectrum disorder and another with severe geriatric major depression) remained nonresponders regardless of the stimulation configuration.

Supporting the specificity of our new targeting method for selectively improving OCD symptoms, the Y-BOCS score improvements were paired with smaller mood and anxiety changes (median BDI and BAI improvements across cohorts 2 and 3 at the last follow-up were 29.9% and 35.5%, respectively). Moreover, unlike previous ALIC DBS trials, we observed no hypomania symptoms (Supplementary Fig. 2). We observed the following transient side effects related to therapeutic DBS settings: restlessness and insomnia ($n = 1$, resolved after lowering the amplitude), tearfulness ($n = 1$, resolved after a few minutes), akathisia and mouth smacking ($n = 1$, resolved after a few minutes), talkativeness ($n = 1$, resolved after a few minutes), anxiety and pins and needles in feet ($n = 1$, resolved after a few minutes), feeling more energetic ($n = 1$, resolved after a few hours), and return of OCD symptoms

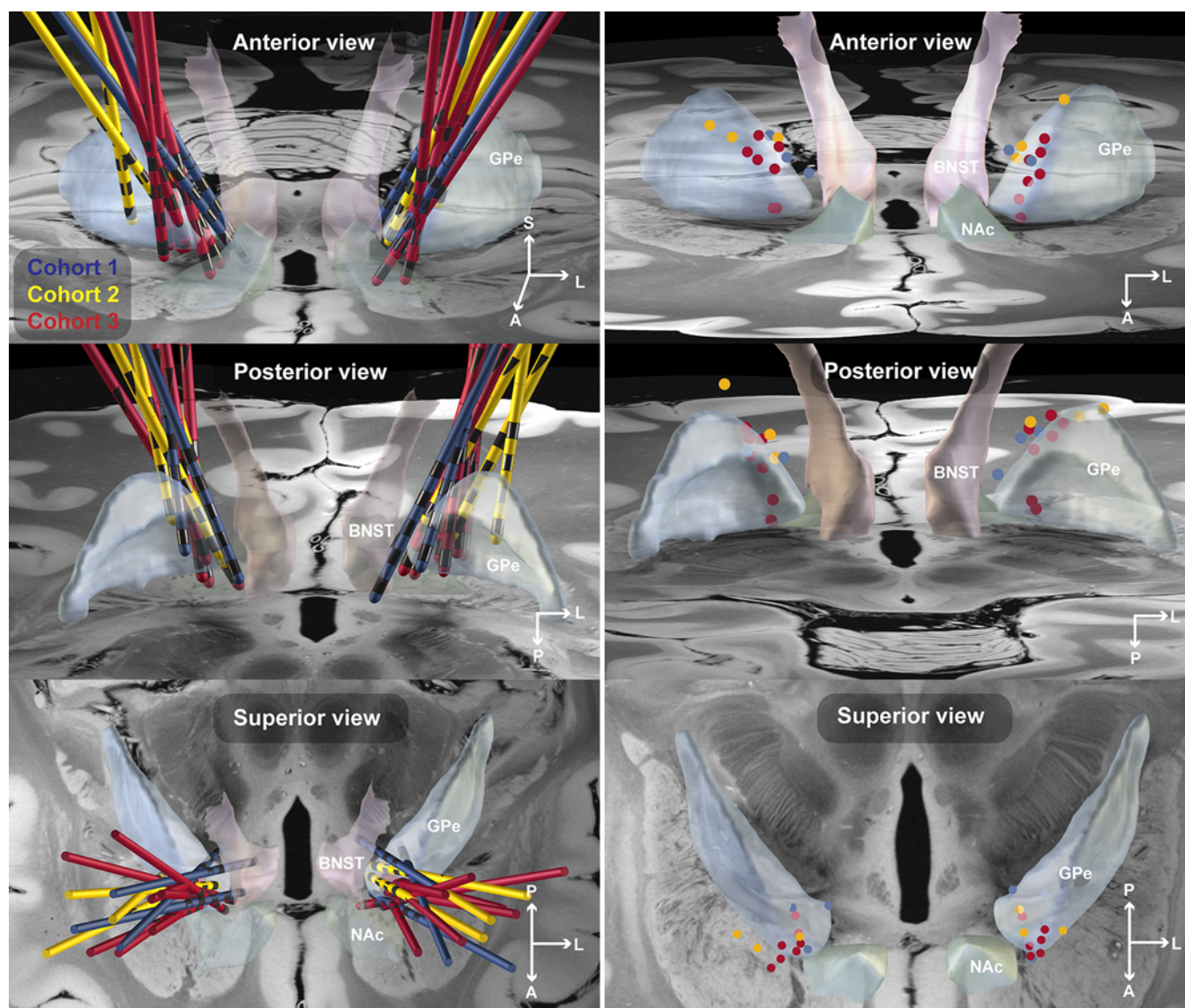


FIG. 4. Electrode location. **Left:** Overview of electrode placement. **Right:** Location of active contacts. BNST = bed nucleus of the stria terminalis; NAc = nucleus accumbens. Figure is available in color online only.

due to device battery depletion ($n = 1$, resolved after battery replacement). One DBS responder (OCD improved by 42%) died of causes unrelated to DBS or OCD.

Tractography-Based Predictive Model

We determined repeated-measures correlations between the activation of individual ALIC pathways and relative reductions in symptom scores (Y-BOCS, BDI, and BAI). After applying the false discovery rate correction for multiple comparisons, stimulation of selective ALIC pathways only statistically significantly correlated with a change in Y-BOCS score but not BDI or BAI score (Fig. 7A). In the left hemisphere, activation of ALIC connections to the vIPFC, thalamus, STN, and the lateral and medial midbrain pathways positively correlated with Y-BOCS score reduction. In the right hemisphere, activation of connections to the vmPFC/OFC, STN, and the lateral

and medial midbrain pathways positively correlated with Y-BOCS score reduction.

To test the predictive value of a model incorporating simultaneous activation of these segregated ALIC pathways, we fitted 3 linear mixed-effect models that predicted Y-BOCS, BDI, and BAI score improvements. The best model fit was for Y-BOCS score (MSE = 753) (Supplementary Results). Despite the large prediction error, there was a significant correlation between the model-predicted and the empirical Y-BOCS scores (repeated-measures correlation coefficient = 0.68, $p < 0.001$) (Fig. 7C). The fitted models for BDI and BAI scores resulted in larger prediction errors (BAI, MSE = 937; BDI, MSE = 7993), so we did not obtain the correlation between model-predicted and empirical BAI and BDI scores.

In the Y-BOCS model, features with positive coefficients represented a clinical improvement, while fea-

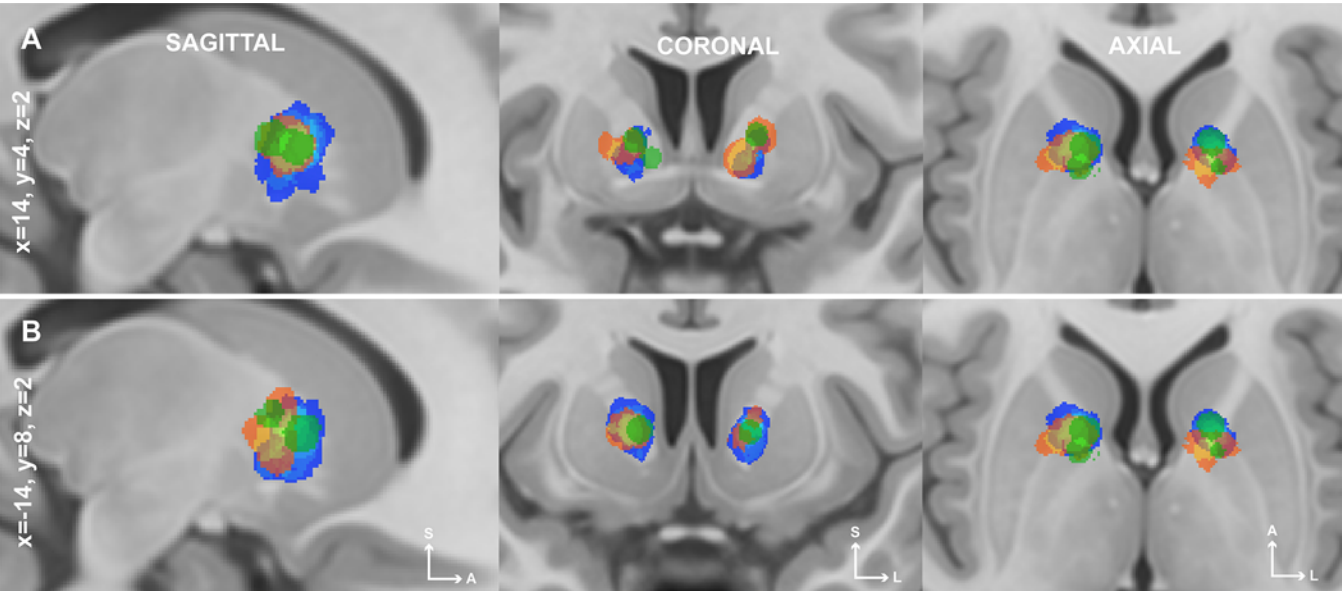


FIG. 5. Aggregated stimulation volumes (N-images) across 3 patient cohorts. Orthogonal brain slices are shown in sagittal (*left*), coronal (*middle*), and axial (*right*) views. **A:** Slices centered at MNI coordinates $x = 14, y = 4, z = 2$ (right hemisphere). **B:** Slices centered at $x = -14, y = 8, z = 2$ (left hemisphere). Color overlays represent stimulation volumes for cohort 1 (*green*), cohort 2 (*yellow*), and cohort 3 (*blue*). The color intensity within each map indicates the number of participants in that cohort with stimulation volumes overlapping each voxel (i.e., N-image values). Brighter regions correspond to higher overlap across patients. Figure is available in color online only.

tures with negative weights had a negative effect on the predicted clinical score (Fig. 7B). Features with positive coefficients in the left and right hemispheres were, in order of importance, left thalamus, right periculate, left

vACC, right STN, left vmPFC/OFC, and right medial midbrain. Features with predictive negative weights were the left periculate, left medial midbrain, and right vmPFC/OFC.

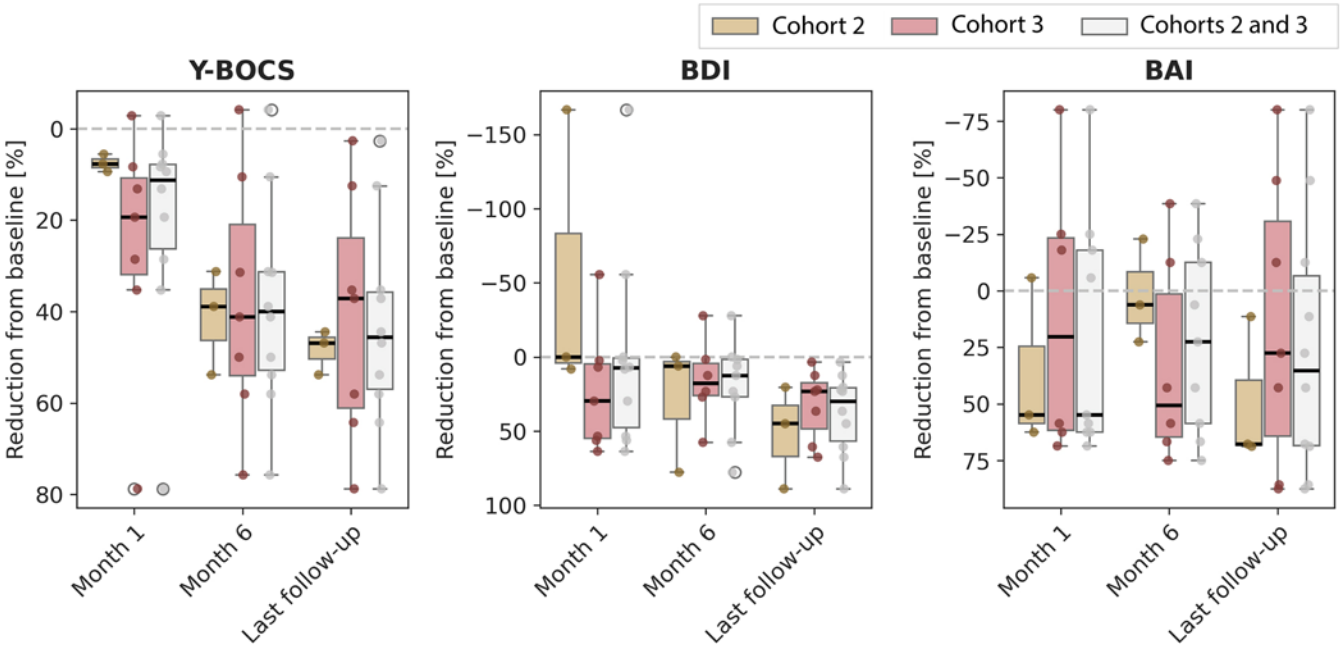


FIG. 6. Clinical symptom scores. Box plots showing the reduction in Y-BOCS (obsessions and compulsions), BDI (depression), and BAI (anxiety) scores from baseline to month 1, month 6, and last follow-up. There were 3 patients in cohort 2, 7 in cohort 3, and 10 in cohorts 2 and 3 combined. Box plots indicate the minimum and maximum values, the interquartile range, and the median of the data per cohort and as a group; individual points correspond to the measurements per patient. Figure is available in color online only.

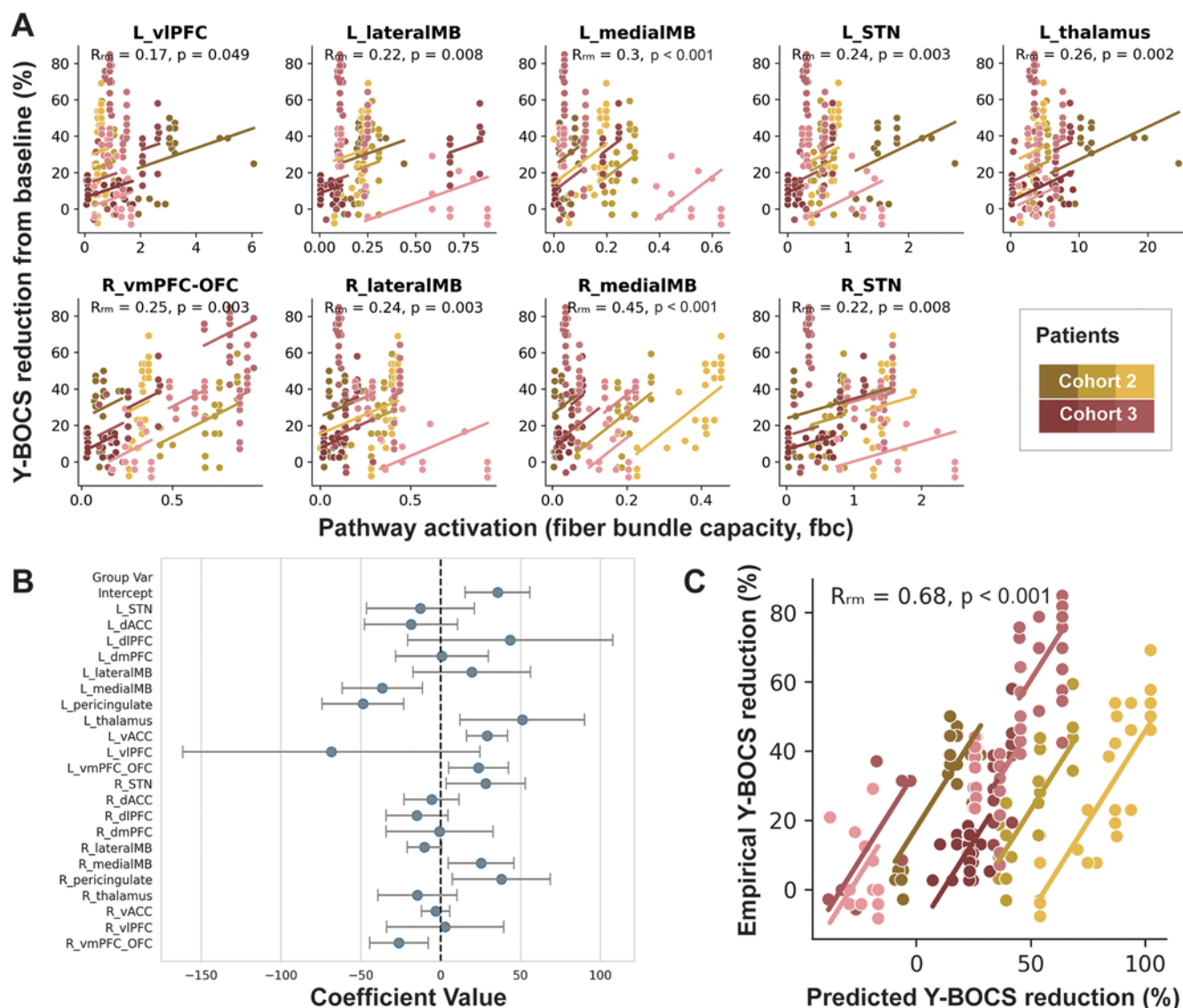


FIG. 7. Tractography-based predictive connectivity model. **A:** Repeated-measures correlation between the activation of the individual ALIC pathways and the percentage of Y-BOCS reduction from baseline. Pathways with significant correlations after false discovery rate correction are shown. The yellow color map indicates individual patients from cohort 2 and the red color map indicates individual patients from cohort 3. **B:** Coefficients in the linear mixed-effects model with Y-BOCS as the outcome variable. The coefficients show the effect, positive or negative, on the predicted Y-BOCS reduction (i.e., clinical improvement or worsening, respectively). Estimates with confidence intervals crossing zero are statistically nonsignificant. **C:** Predicted Y-BOCS reduction versus empirical values. MB = midbrain; R_{rm} = repeated-measures correlation coefficient. Figure is available in color online only.

Updated Common Responder Map

Finally, we generated an updated common map using high-resolution 7T imaging data from the 8 responders from cohorts 2 and 3 (Fig. 8). The updated map agreed with the initial one: at the cortical level, connections to the vIPFC and vmPFC/OFC, and subcortically, a subset of streamlines entering the thalamus (Fig. 8A) and a subset descending toward the STN and brainstem (Fig. 8C).

Discussion

This study reports an evolution toward precision ALIC DBS targeting based on patient-specific tractography for a

fast, consistent, and specific reduction of OCD symptoms. Our tractography-based stimulation model selectively predicted the improvement in obsessions and compulsions rather than a more general effect on mood or anxiety. If confirmed in larger prospective cohorts, our method might allow for a more predictable and implementable tractography-based target for OCD DBS.

To our knowledge, this is the first study generating and prospectively utilizing a common responder map of ALIC DBS. Most OCD DBS connectomic studies have only retrospectively used normative connectome or individual tractography data to analyze the relationship between white

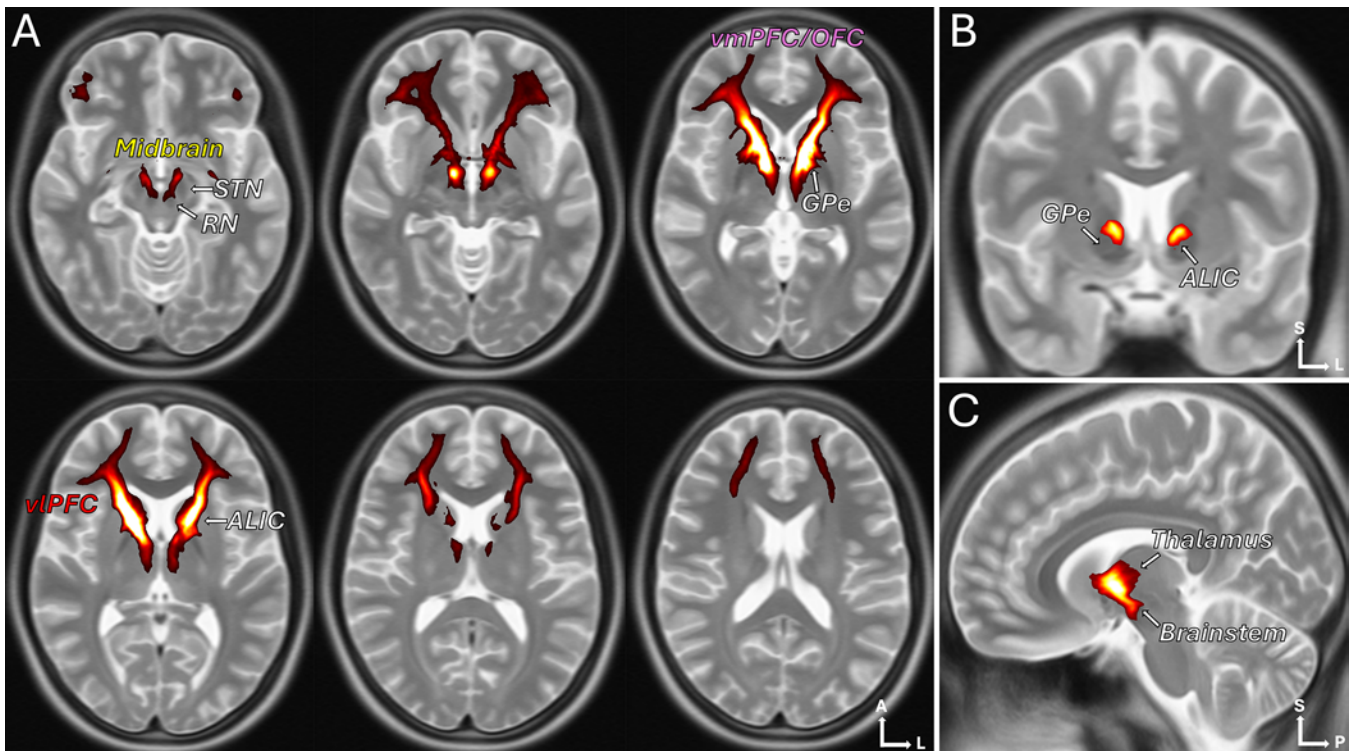


FIG. 8. Updated common responder map. Axial (A), coronal (B), and sagittal (C) average tract-density images derived from the streamlines included in the stimulation volumes of the responder patients in cohorts 2 and 3. Brighter colors in the heat map represent a higher density of streamlines' weights. Figure is available in color online only.

matter activation and clinical outcome.^{12,14,21,28,29} Other studies have used patient-specific tractography to target individual white matter tracts (e.g., superolateral medial forebrain bundle).^{15,30} Our data-driven approach did not restrict the connectivity to a subset of bundles. Instead, the new empirically derived target map highlighted white matter connections from the stimulation sites of responder patients to the vlPFC, vmPFC/OFC, and specific midbrain regions.

Our responder map agrees with established cortical projection maps from the ALIC.⁹ Subcortically, our midbrain connections align with the lateral and medial fiber depictions at the ventral tegmental area observed in prior anatomical studies.^{10,31,32} Our lateral midbrain connections to the STN may correspond with the hyperdirect pathway described in normative ALIC DBS analyses.^{10,14}

After defining the therapeutic responder group map, we constructed brain masks within this map and registered these to the patient's native space by combining linear and nonlinear transformation for target planning. These disease-specific waypoint masks allowed us to extract patient-specific white matter bundles that project to vlPFC, vmPFC/OFC, and midbrain. The optimal target was then selected by maximally engaging these extracted patient-specific white matter bundles. The disease-specific waypoint masks can be easily shared and applied to validate the current tractography-based targeting method for ALIC OCD DBS in other clinical centers. Furthermore, ongoing data collection in new OCD DBS responders will update our responder map and disease-specific waypoint masks over time.

Our new DBS target appears to result in more OCD symptom-specific benefits than previous approaches. Conventional ventral capsule/ventral striatum DBS immediately improves mood and anxiety,^{5,33,34} often with initial hypomania,³⁵ prior to and independent of a more gradual improvement in obsessions and compulsions.³⁶ Stimulation of our single-contact tractography-based target more consistently improved obsessions and compulsions without dominant anxiety or mood changes or hypomania, and our stimulation model predicted improvement in obsessions and compulsions but not depression or anxiety. A recent retrospective multicenter analysis also suggested that the optimal ALIC stimulation location for OCD improvement differed from the optimal sites for the improvement of depression and anxiety.¹⁶ Our correlation analysis indicated that the sweet spot of therapeutic stimulation, located in the ventral ALIC near the GPe, activated thalamic and midbrain fibers connected to the vlPFC and vmPFC/OFC. These connections have been implicated in OCD-specific impairments of cognitive flexibility and inhibitory control.^{7,37} Our GPe network node is part of the indirect associative corticostriatal pathway where inhibitory high-frequency DBS may help overcome inflexible thoughts and behaviors.³⁷ In agreement, OCD models suggest hyperactivity in GPe-prefrontal indirect pathways that can be reduced by optogenetic inhibition.³⁸ The anteromedial STN is another node in this indirect associative pathway, and DBS in this region also improves OCD and cognitive flexibility more than mood.^{39,40} Our initial common responder map included subcortical connections passing

by the medial STN, confirmed in the predictive connectivity model and the updated common map.

Neither the responder common map nor the predictive model suggested that connectivity to the dACC was strongly involved in OCD improvements, although the dACC has been implicated in the pathophysiology of OCD⁴¹ and was involved in a normative connectome OCD DBS tract.¹⁴ Activation of the dACC has been described from electrodes placed in the dorsomedial ALIC,⁹ whereas our stimulation site was more ventral and lateral to this location. However, the lack of activation of dACC fibers could also be due to differences in the tractography methodology of our model and analyses performed with normative connectomes.

Limitations

Our study was not designed as a sham-controlled efficacy trial. We use this case series with a limited sample size to present the evolution of our tractography-based targeting method. DBS for OCD is not yet a standard treatment, leading to relatively small patient populations at individual centers. However, combining patient-specific imaging data from multiple centers presents significant challenges, so normative connectome data is typically used.

Complementary to normative connectome data, patient-specific tractography can provide improved insights into individual connectivity variance. However, imaging quality and tractography methods have an impact on the output.⁴² We used 7T high-resolution multishell diffusion MRI and defined anatomical priors^{8,9} to improve the accuracy of the tractography reconstructions.^{43,44}

Other limitations apply to the current work. First, the patients included in the study had different follow-up times, and some patients in cohort 3, with shorter follow-ups than those in cohort 2, may experience further long-term beneficial effects. Second, we implemented simplified estimations of stimulation volumes. Complex axonal models⁴⁵ are computationally demanding and impractical in clinical settings. Thus, our method is a viable solution to generate patient-specific stimulation models. Finally, the Y-BOCS outcome predictions of our linear-mixed effects model were significantly correlated to the empirical scores, although the prediction errors cannot be disregarded. Therefore, the model should be further validated with different datasets and larger cohorts of patients.

Conclusions

This study demonstrates that patient-specific tractography can help precisely select surgical targets for ALIC DBS. This approach leads to a consistent and selective reduction of OCD symptoms with minimal parameter adjustments. Our therapeutic map of ALIC connections, when validated, can help overcome the variability and unpredictability of DBS for OCD and contribute to a more widespread clinical implementation.

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Disclosures

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Supplemental Information

Online-Only Content

Supplemental material is available with the online version of the article.

Supplementary Information. <https://thejns.org/doi/suppl/10.3171/2025.6.JNS243066>.

Data Availability

Pipelines for diffusion MRI processing and data analysis are available at <https://gitlab.switch.ch/alba.seguraamil/alicconnectivity.git>.

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